Application of Information Theory, Lecture 2 [I]
Joint & Conditional Entropy, Mutual Information

Handout Mode

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October 20, November 03, 2026

[I] Prepared: 2026/11/03,14:55:28.

Part I

Joint and Conditional Entropy

Joint entropy

- Recall that the entropy of rv X is defined by

$$\mathrm{H}(X) = -\sum_{x \in \mathcal{X}} \mathrm{P}_X(x) \log \mathrm{P}_X(x) = \mathop{\mathrm{E}}_{x \overset{R}{\leftarrow} X} [\mathrm{H}_X(x)]$$

- Shorter notation: for $X \sim p$, let $\mathrm{H}(X) = -\sum_x p(x) \log p(x)$ (where the summation is over the domain of X).
- The joint entropy of (jointly distributed) rvs X and Y with $(X, Y) \sim p$, is

$$\mathrm{H}(X, Y) = -\sum_{x,y} p(x, y) \log p(x, y) = \mathop{\mathrm{E}}_{(x,y) \overset{R}{\leftarrow} (X,Y)} [\mathrm{H}_{X,Y}(x, y)]$$

  This is simply the entropy of the rv $Z = (X, Y)$.

- Example:

$X \backslash Y$	0	1
0	$\frac{1}{4}$	$\frac{1}{4}$
1	$\frac{1}{2}$	0

$$\begin{aligned}\mathrm{H}(X, Y) &= \frac{1}{2} \log \frac{1}{\frac{1}{2}} + \frac{1}{4} \log \frac{1}{\frac{1}{4}} + \frac{1}{4} \log \frac{1}{\frac{1}{4}} \\ &= \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 2 = 1\frac{1}{2}\end{aligned}$$

Joint entropy, cont.

- The joint entropy of $(X_1, \ldots, X_n) \sim p$, is

$$\mathrm{H}(X_1, \ldots, X_n) = -\sum_{x_1, \ldots, x_n} p(x_1, \ldots, x_n) \log p(x_1, \ldots, x_n)$$

Conditional entropy

- Let $(X, Y) \sim p$, let $p_X(x) = \sum_y p(x, y)$, $p_Y(y) = \sum_x p(x, y)$ and $p_{Y|X}(y|x) = \frac{p(x,y)}{p_X(x)}$.
- Conditional sample entropy: $\mathrm{H}_{Y|X}(y|x) = -\log p_{Y|X}(y|x)$
- For $x \in \mathrm{Supp}(X)$, the random variable $Y|_{X=x}$ is well defined (distributed according to $q(y) = p_{Y|X}(y|x)$).
- The entropy of Y conditioned on X is defined by

$$\mathrm{H}(Y|X) := \mathop{\mathrm{E}}_{x \xleftarrow{R} X} [\mathrm{H}(Y|_{X=x})] = \mathop{\mathrm{E}}_{(x,y) \xleftarrow{R} (X,Y)} \left[\mathrm{H}_{Y|X}(y|x)\right]$$

- Measures the uncertainty in Y given X.
- $$\begin{aligned}\mathrm{H}(Y|X) &= \mathop{\mathrm{E}}_{(x,y) \xleftarrow{R} (X,Y)} \left[\mathrm{H}_{Y|X}(y|x)\right] = -\mathop{\mathrm{E}}_{(x,y) \xleftarrow{R} (X,Y)} \left[\log p_{Y|X}(y|x)\right] \\ &= -\mathop{\mathrm{E}}_{Z = p_{Y|X}(Y|X)} [\log Z]\end{aligned}$$

Conditional entropy, cont.

- Example

$X \backslash Y$	0	1
0	$\frac{1}{4}$	$\frac{1}{4}$
1	$\frac{1}{2}$	0

What is $H(Y|X)$ and $H(X|Y)$?

$$\begin{aligned} H(Y|X) &= \mathop{E}_{x \overset{R}{\leftarrow} X}[H(Y|_{X=x})] \\ &= \frac{1}{2} H(Y|_{X=0}) + \frac{1}{2} H(Y|_{X=1}) \\ &= \frac{1}{2} H(\frac{1}{2}, \frac{1}{2}) + \frac{1}{2} H(1,0) = \frac{1}{2}. \end{aligned}$$

$$\begin{aligned} H(X|Y) &= \mathop{E}_{y \overset{R}{\leftarrow} Y}[H(X|_{Y=y})] \\ &= \frac{3}{4} H(X|_{Y=0}) + \frac{1}{4} H(X|_{Y=1}) \\ &= \frac{3}{4} H(\frac{1}{3}, \frac{2}{3}) + \frac{1}{4} H(1,0) = 0.6887 \neq H(Y|X). \end{aligned}$$

Conditional entropy, cont..

- $$\begin{aligned}
\mathrm{H}(X|Y,Z) &= \mathop{\mathrm{E}}_{(y,z)\overset{R}{\leftarrow}(Y,Z)}[\mathrm{H}(X|_{Y=y,Z=z})] \\
&= \mathop{\mathrm{E}}_{y\overset{R}{\leftarrow}Y}\left[\mathop{\mathrm{E}}_{z\overset{R}{\leftarrow}Z|_{Y=y}}[\mathrm{H}(X|_{Y=y,Z=z})]\right] \\
&= \mathop{\mathrm{E}}_{y\overset{R}{\leftarrow}Y}\left[\mathop{\mathrm{E}}_{z\overset{R}{\leftarrow}Z|_{Y=y}}[\mathrm{H}((X|_{Y=y})|_{Z=z})]\right]
\end{aligned}$$

Let $(X_y, Z_y) = (X,Z)|_{Y=y}$. Then

$$\begin{aligned}
\mathrm{H}(X|Y,Z) &= \mathop{\mathrm{E}}_{y\overset{R}{\leftarrow}Y}\left[\mathop{\mathrm{E}}_{z\overset{R}{\leftarrow}Z_y}[\mathrm{H}(X_y|_{Z=z})]\right] \\
&= \mathop{\mathrm{E}}_{y\overset{R}{\leftarrow}Y}\left[\mathop{\mathrm{E}}_{z\overset{R}{\leftarrow}Z_y}\left[\mathrm{H}(X_y|_{Z_y=z})\right]\right]
\end{aligned}$$

Relating joint entropy to conditional entropy

- What is the relation between $H(X)$, $H(Y)$, $H(X,Y)$ and $H(Y|X)$?
- Intuitively, $0 \leq H(Y|X) \leq H(Y)$

  Non-negativity is immediate. We prove the upper bound later.
- We will also see that $H(Y|X) = H(Y)$ iff X and Y are independent.
- In our example, $H(Y) = H(\frac{3}{4}, \frac{1}{4}) > \frac{1}{2} = H(Y|X)$
- Note that $H(Y|_{X=x})$ might be larger than $H(Y)$ for some $x \in \text{Supp}(X)$.
- Chain rule (proved next). $H(X,Y) = H(X) + H(Y|X)$
- Intuitively, uncertainty in (X,Y) is the uncertainty in X plus the uncertainty in Y given X.
- $H(Y|X) = H(X,Y) - H(X)$ is an alternative definition for $H(Y|X)$.

Chain rule (for the entropy function)

Claim 1

For rvs X, Y, it holds that $\mathrm{H}(X, Y) = \mathrm{H}(X) + \mathrm{H}(Y|X)$.

- Proof immediately follows from the grouping axiom:

$X \backslash Y$			
	$P_{1,1}$	$\cdots$	$P_{1,n}$
	$\vdots$	$\vdots$	$\vdots$
	$P_{n,1}$	$\cdots$	$P_{n,n}$

Let $q_i = \sum_{j=1}^{n} P_{i,j}$ $\quad (= \Pr[X = i])$

$$\begin{aligned}
&\mathrm{H}(P_{1,1}, \ldots, P_{n,n}) \\
&= \mathrm{H}(q_1, \ldots, q_n) + \sum_i q_i \, \mathrm{H}\left(\frac{P_{i,1}}{q_i}, \ldots, \frac{P_{i,n}}{q_i}\right) \\
&= \mathrm{H}(X) + \mathrm{H}(Y|X).
\end{aligned}$$

- Another proof. Let $(X, Y) \sim p$, and recall that $p(x, y) = p_X(x) \cdot p_{Y|X}(y|x)$.

$\Longrightarrow$ $\log p(x, y) = \log p_X(x) + \log p_{Y|X}(y|x)$

$\Longrightarrow$ $\mathrm{E}\left[\log p(X, Y)\right] = \mathrm{E}\left[\log p_X(X)\right] + \mathrm{E}\left[\log p_{Y|X}(Y|X)\right]$

$\Longrightarrow$ $\mathrm{H}(X, Y) = \mathrm{H}(X) + \mathrm{H}(Y|X)$.

$H(Y|X) \leq H(Y)$

Jensen inequality: for any concave function f, values $t_1, \ldots, t_k$ and $\lambda_1, \ldots, \lambda_k \in [0,1]$ with $\sum_i \lambda_i = 1$, it holds that $\sum_i \lambda_i f(t_i) \leq f(\sum_i \lambda_i t_i)$.
Let $(X, Y) \sim p$.

$$
\begin{aligned}
H(Y|X) &= -\sum_{x,y} p(x,y) \log p_{Y|X}(y|x) \\
&= \sum_{x,y} p(x,y) \log \frac{p_X(x)}{p(x,y)} \\
&= \sum_{x,y} p_Y(y) \cdot \frac{p(x,y)}{p_Y(y)} \log \frac{p_X(x)}{p(x,y)} \\
&= \sum_{y} p_Y(y) \sum_{x} \frac{p(x,y)}{p_Y(y)} \log \frac{p_X(x)}{p(x,y)} \\
&\leq \sum_{y} p_Y(y) \log \sum_{x} \frac{p(x,y)}{p_Y(y)} \frac{p_X(x)}{p(x,y)} \\
&= \sum_{y} p_Y(y) \log \frac{1}{p_Y(y)} = H(Y).
\end{aligned}
$$

$H(Y|X) \leq H(Y)$ cont.

- Assume X and Y are independent (i.e., $p(x,y) = p_X(x) \cdot p_Y(y)$ for any x, y)

$\Longrightarrow$ $p_{Y|X}(y|x) = p_Y(y)$ for any x, y

$\Longrightarrow$ $H(Y|X) = H(Y)$

- Is the converse also true: $H(Y|X) = H(Y)$ implies X and Y are independent?

  Yes, since log is strictly concave in the range. Equality happens iff all t_i (in Jensen inequality) are the same,

- which happens iff $p(x,y) = p_X(x)p_Y(y)$ for all x, y

Other inequalities

- $\mathrm{H}(X), \mathrm{H}(Y) \leq \mathrm{H}(X, Y) \leq \mathrm{H}(X) + \mathrm{H}(Y)$.
  Follows from $\mathrm{H}(X, Y) = \mathrm{H}(X) + \mathrm{H}(Y|X)$.
  - Left inequality since $\mathrm{H}(Y|X)$ is non-negative.
  - Right inequality since $\mathrm{H}(Y|X) \leq \mathrm{H}(Y)$.
- $\mathrm{H}(X, Y|Z) = \mathrm{H}(X|Z) + \mathrm{H}(Y|X, Z)$ (by chain rule)
- $\mathrm{H}(X|Y, Z) \leq \mathrm{H}(X|Y)$

Proof:

$$\begin{aligned}
\mathrm{H}(X|Y,Z) &= \mathop{\mathrm{E}}_{(y,z)\overset{R}{\leftarrow}(Y,Z)} \left[\mathrm{H}(X|_{(Y,Z)=(y,z)})\right] \\
&= \mathop{\mathrm{E}}_{y\overset{R}{\leftarrow}Y} \left[\mathop{\mathrm{E}}_{z\overset{R}{\leftarrow}Z|_{Y=y}} \left[\mathrm{H}(X|_{(Y,Z)=(y,z)})\right] \right] \\
&= \mathop{\mathrm{E}}_{y\overset{R}{\leftarrow}Y} \left[\mathop{\mathrm{E}}_{z\overset{R}{\leftarrow}Z|_{Y=y}} \left[\mathrm{H}((X|_{Y=y})|_{Z=z})\right] \right] \\
&= \mathop{\mathrm{E}}_{y\overset{R}{\leftarrow}Y} \left[\mathrm{H}(X|_{Y=y}|Z|_{Y=y})\right] \\
&\leq \mathop{\mathrm{E}}_{y\overset{R}{\leftarrow}Y} \left[\mathrm{H}(X|_{Y=y})\right]
\end{aligned}$$

Chain rule (for the entropy function), general case

Claim 2

For rvs $X_1, \ldots, X_k$, it holds that
$H(X_1, \ldots, X_k) = H(X_1) + H(X_2|X_1) + \ldots + H(X_k|X_1, \ldots, X_{k-1})$.

Proof: ?

- Extremely useful property!
- Analogously to the two-variable case, it also holds that:
- $H(X_i) \leq H(X_1, \ldots, X_k) \leq \sum_i H(X_i)$
- $H(X_1, \ldots, X_k|Y) \leq \sum_i H(X_i|Y)$

Examples

- (from last class) Let $X_1, \ldots, X_n$ be Boolean iid with $X_i \sim (\frac{1}{3}, \frac{2}{3})$. Compute $\mathrm{H}(X_1, \ldots, X_n)$
- As above, but X_n is set to $\bigoplus_{1 \leq i \leq n-1} X_i$?
  - Via chain rule?
  - Via mapping?

Summary: equalities and inequalities

- $0 \leq \mathrm{H}(X) \leq \log |\mathrm{Supp}(X)|$
- $\mathrm{H}(Y|X) \leq \mathrm{H}(Y)$ (equality if independent)
- $\mathrm{H}(X), \mathrm{H}(Y) \leq \mathrm{H}(X, Y) \leq \mathrm{H}(X) + \mathrm{H}(Y)$ (rhs is equality if independent).
- Many other equalities and inequalities, develop when you need them

Concavity of entropy function

Let $p = (p_1, \ldots, p_n)$ and $q = (q_1, \ldots, q_n)$ be two distributions, and for $\lambda \in [0, 1]$ consider the distribution $\tau_\lambda = \lambda p + (1 - \lambda) q$.
(i.e., $\tau_\lambda = (\lambda p_1 + (1 - \lambda) q_1, \ldots, \lambda p_n + (1 - \lambda) q_n)$).

Claim 3

$\mathrm{H}(\tau_\lambda) \geq \lambda \mathrm{H}(p) + (1 - \lambda) \mathrm{H}(q)$

Proof:

- Let Y over $\{0, 1\}$ be 0 wp λ
- Let X be distributed according to p if $Y = 0$ and according to q otherwise.
- $\mathrm{H}(\tau_\lambda) = \mathrm{H}(X) \geq \mathrm{H}(X \mid Y) = \lambda \mathrm{H}(p) + (1 - \lambda) \mathrm{H}(q)$

We are now certain that we drew the graph of the (two-dimensional) entropy function right...

Section 1

Applications

Extractors

- Let $X_1, \ldots, X_n$ be Boolean iids with $X_i \sim (p, 1-p)$ and let $X = X_1, \ldots, X_n$. Let f be such that $\Pr[f(X) = z \mid |f(X)| = k] = \Pr[f(X) = z' \mid |f(X)| = k]$, for every $k \in \mathbb{N}$ and $z, z' \in \{0,1\}^k$. Let $K = |f(X)|$.

  Prove that $\mathrm{E}[K] \leq n \cdot h(p)$.

- $$\begin{aligned} n \cdot h(p) &= \mathrm{H}(X_1, \ldots, X_n) \\ &\geq \mathrm{H}(f(X), K) \\ &= \mathrm{H}(K) + \mathrm{H}(f(X) \mid K) \\ &= \mathrm{H}(K) + \mathrm{E}[K] \\ &\geq \mathrm{E}[K] \end{aligned}$$

- Interpretation

  How many bits is possible to extract from *bit-fixing source*

- Upper bounds

  Bit fixing source extractors, e.g., von newman extractors.

Sorting

- How many comparisons does it take to sort n elements?

  Let S be a sorter for n elements making t comparisons.

  What can we say about t?

- Let X be a uniform random permutation of $[n]$ and let $Y_1, \ldots, Y_t$ be the answers S gets when sorting X
- X is determined by $Y_1, \ldots, Y_t$

  Namely, $X = f(Y_1, \ldots, Y_t)$ for some function f
- $\mathrm{H}(X) = \log(n!)$
- $$\begin{aligned} \mathrm{H}(X) &= \mathrm{H}(f(Y_1, \ldots, Y_t)) \\ &\leq \mathrm{H}(Y_1, \ldots, Y_t) \\ &\leq \sum_i \mathrm{H}(Y_i) \\ &\leq t \end{aligned}$$

$\Longrightarrow \quad t \geq \log n! = \Theta(n \log n)$

Graph Covering

- How many graphs of certain type it takes to cover the full graph?
- K_n — the complete graph over $[n]$
- Let $G_1, \ldots, G_t$ be bipartite graphs over $[n]$ with $\bigcup_i G_i = K_n$.
  What can we say about t?
- Clearly, $t \geq \frac{\binom{n}{2}}{(n/2)^2} \approx 2$, but can we give a better bound?

Theorem 4

Let $G_1, \ldots, G_t$ be bipartite graphs over $[n]$ with $\bigcup_{i=1}^{t} G_i = K_n$, then $t \geq \log n$.

Proof: Let $\chi(G)$ be the chromatic number of G.

- $\chi(G_i) \leq 2$ and $\chi(K_n) = n$.
- $\chi(G \cup G') \leq \chi(G) \cdot \chi(G')$.(?)

$\Longrightarrow$ $\chi(\bigcup_{i=1}^{t} G_i) \leq 2^t$

$\Longrightarrow$ $t \geq \log n$

Proving Thm 4 using entropy

- $G_i = (A_i, B_i, E_i)$
- $X \overset{R}{\leftarrow} [n]$
- $Y_i = \begin{cases} 0, & X \in A_i \\ 1, & X \in B_i \end{cases}$
- X is determined by $Y_1, \ldots, Y_t$ (?)

$$\begin{aligned} 0 = \mathrm{H}(X \mid Y_1, \ldots, Y_t) &= \mathrm{H}(X, Y_1, \ldots, Y_t) - \mathrm{H}(Y_1, \ldots, Y_t) \\ &\geq \mathrm{H}(X) - \sum_i \mathrm{H}(Y_i) \\ &\geq \log n - t. \end{aligned}$$

Extensions

- $\text{nonIs}(G)$ — non-isolated vertices in G.

Theorem 5

Let $G_1, \ldots, G_t$ be bipartite graphs over $[n]$ with $\bigcup_{i=1}^{t} G_i = K_n$, then $\frac{1}{n}\sum_{i=1}^{t} |\text{nonIs}(G_i)| \geq \log n$.

Definition 6 (Graph content)

Let G be a graph over $[n]$, let $Z \overset{R}{\leftarrow} \text{nonIs}(G)$ and let $\hat{\chi}$ be a (valid) coloring of G such that $\text{H}(\hat{\chi}(Z))$ is minimal. Then $\text{content}(G) := \frac{|\text{nonIs}(G)|}{n} \cdot \text{H}(\hat{\chi}(Z))$.

Theorem 7

Let $G_1, \ldots, G_t$ be graphs over $[n]$ with $\bigcup_{i=1}^{t} G_i = K_n$. Then $\sum_i \text{content}(G_i) \geq \log n$.

- Since $\text{content}(G) \leq \frac{|\text{nonIs}(G)|}{n}$ for bipartite G, Thm 7 yields Thm 5.

Proving Thm 7

- Let $\chi_1, \ldots, \chi_t$ be (valid) coloring of $G_1, \ldots, G_t$.
- Let $X \overset{R}{\leftarrow} [n]$, and let
  $$Y_i = \begin{cases} \chi_i(X) & X \in \text{nonIs}(G_i) \\ \chi_i(Z_i) & \text{otherwise, for } Z_i \overset{R}{\leftarrow} \text{nonIs}(G_i) \text{ (ind. of the other } Z\text{'s).} \end{cases}$$
- X is determined by $Y_1, \ldots, Y_t$ (?)
  $$\begin{aligned} 0 = \mathrm{H}(X|Y_1, \ldots, Y_t) &= \mathrm{H}(X, Y_1, \ldots, Y_t) - \mathrm{H}(Y_1, \ldots, Y_t) \\ &\geq \mathrm{H}(X) + \mathrm{H}(Y_1, \ldots, Y_t|X) - \sum_i \mathrm{H}(Y_i) \\ &= \log n + \mathrm{H}(Y_1, \ldots, Y_t|X) - \sum_i \mathrm{H}(Y_i). \end{aligned}$$
- $Y_1, \ldots, Y_t$ are independent conditioned on X —
  $\Pr[Y_1 = y_1 \wedge Y_2 = y_2 \mid X = x] = \Pr[Y_1 = y_1 \mid X = x] \cdot \Pr[Y_2 = y_2 \mid X = x]$

$\Longrightarrow$ $\mathrm{H}(Y_1, \ldots, Y_t|X) = \sum_i \mathrm{H}(Y_i|X)$ (board)

$\Longrightarrow$ $\sum_i \mathrm{H}(Y_i) - \sum_i \mathrm{H}(Y_i|X) \geq \log n$

- Since $\mathrm{H}(Y_i) = \mathrm{H}(\chi_i(Z_i))$ and $\mathrm{H}(Y_i|X) = (1 - \frac{|\text{nonIs}(G_i)|}{n}) \cdot \mathrm{H}(\chi_i(Z_i))$,
  it follows that $\sum_i \mathrm{H}(\chi_i(Z_i)) \frac{|\text{nonIs}(G_i)|}{n} \geq \log n$. $\square$

Extension

Let $\alpha(G)$ be the size of the maximal independent set in G.

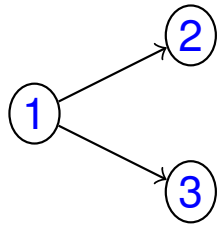
Theorem 8

Let $G, G_1, \ldots, G_t$ be graphs over $[n]$ with $\bigcup_{i=1}^{t} G_i = G$, then $\sum \text{content}(G_i) \geq \log \frac{n}{\alpha(G)}$.

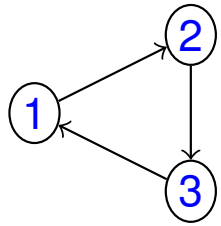
Proof: HW

Counting # of graph homomorphisms

- $T = (V_T, E_T)$ — directed graph (no self loops)
- $G = (V_G, E_G)$



- $H = (V_H, E_H)$



- (x_1, x_2, x_3) is an homomorphism of G in T, if $x_1, x_2, x_3 \in V_T$ and $(i, j) \in E_G \implies (x_i, x_j) \in E_T$ (might be $x_1 = x_2$)
- Example: see board
- $\mathrm{Hom}(X, T)$: all homomorphisms of X in T
- Claim: $|\mathrm{Hom}(H, T)| \leq |\mathrm{Hom}(G, T)|$
- Trivial if G would be a subgraph of H
- Special case of a more general theorem

Proving the claim

- $(X_1, X_2, X_3) \overset{R}{\leftarrow} \mathrm{Hom}(H, T)$
- $$\begin{aligned}\log|\mathrm{Hom}(H,T)| &= \mathrm{H}(X_1,X_2,X_3)\\ &= \mathrm{H}(X_1)+\mathrm{H}(X_2|X_1)+\mathrm{H}(X_3|X_1,X_2)\\ &\le \mathrm{H}(X_1)+\mathrm{H}(X_2|X_1)+\mathrm{H}(X_3|X_2)\\ &= \mathrm{H}(X_1)+2\cdot\mathrm{H}(X_2|X_1) \qquad \text{(by symmetry of } H\text{)}\end{aligned}$$

- Let $D_2(x)$ be the distribution of $X_2|X_1 = x$, and let $X_2' \sim D_2(X_1)$
- $$\begin{aligned}\mathrm{H}(X_1,X_2,X_2') &= \mathrm{H}(X_1)+\mathrm{H}(X_2|X_1)+\mathrm{H}(X_2'|X_1,X_2)\\ &= \mathrm{H}(X_1)+\mathrm{H}(X_2|X_1)+\mathrm{H}(X_2'|X_1)\\ &= \mathrm{H}(X_1)+2\cdot\mathrm{H}(X_2|X_1)\\ &= \log|\mathrm{Hom}(H,T)|\end{aligned}$$
- $(x_1, x_2), (x_1, x_2') \in E_T \implies (x_1, x_2, x_2') \in \mathrm{Hom}(G, T)$

$\implies (X_1, X_2, X_2') \in \mathrm{Hom}(G, T)$

$\implies \mathrm{H}(X_1, X_2, X_2') \le \log|\mathrm{Hom}(G, T)|$

$\implies \log|\mathrm{Hom}(H, T)| \le \log|\mathrm{Hom}(G, T)|$. $\square$

Part II

Mutual Information

Mutual information

- $I(X;Y)$ — the "information" that X gives on Y
- $$\begin{aligned} I(X;Y) &:= \mathrm{H}(Y) - \mathrm{H}(Y|X) \\ &= \mathrm{H}(Y) - (\mathrm{H}(X,Y) - \mathrm{H}(X)) \\ &= \mathrm{H}(X) + \mathrm{H}(Y) - \mathrm{H}(X,Y) \\ &= \mathrm{H}(X) - \mathrm{H}(X|Y) \\ &= I(Y;X). \end{aligned}$$
- The mutual information that X gives about Y equals the mutual information that Y gives about X.
- $I(X;Y) \geq 0$. When 0?
- $I(X;X) = \mathrm{H}(X)$
- $I(X;f(X)) = \mathrm{H}(f(X))$ (and smaller than $\mathrm{H}(X)$ if f is non-injective)
- $I(X;Y,Z) \geq I(X;Y), I(X;Z)$ (since $\mathrm{H}(X \mid Y,Z) \leq \mathrm{H}(X \mid Y), \mathrm{H}(X \mid Z)$)
- $I(X;Y|Z) := \mathrm{H}(Y|Z) - \mathrm{H}(Y|X,Z)$ ≥ 0
- $I(X;Y|Z) = I(Y;X|Z)$ (since $I(X';Y') = I(Y';X')$)

Numerical example

- Example

$X \backslash Y$	0	1
0	$\frac{1}{4}$	$\frac{1}{4}$
1	$\frac{1}{2}$	0

$$
\begin{aligned}
I(X;Y) &= \mathrm{H}(X) - \mathrm{H}(X|Y) \\
&= 1 - \frac{3}{4} \cdot h(\frac{1}{3}) \\
&= I(Y;X) \\
&= \mathrm{H}(Y) - \mathrm{H}(Y|X) \\
&= h(\frac{1}{4}) - \frac{1}{2} h(\frac{1}{2})
\end{aligned}
$$

Chain rule for mutual information

Claim 9 (Chain rule for mutual information)

For rvs $X_1, \ldots, X_k, Y$, it holds that
$I(X_1, \ldots, X_k; Y) = I(X_1; Y) + I(X_2; Y|X_1) + \ldots + I(X_k; Y|X_1, \ldots, X_{k-1})$.

Proof: ? HW

Examples

- Let $X_1, \ldots, X_{n-1}$ be iid uniform bits (i.e., $X_i \sim (\frac{1}{2}, \frac{1}{2})$), and let $X_n = \bigoplus_{i \in [n-1]} X_i$. Compute $I(X_1, \ldots, X_{n-1}; X_n)$.
  - Directly,
    $I(X_1, \ldots, X_{n-1}; X_n) = \mathrm{H}(X_n) - \mathrm{H}(X_n | X_1, \ldots, X_{n-1}) = 1 - 0 = 1$
  - Using chain rule,

$$\begin{aligned} & I(X_1, \ldots, X_{n-1}; X_n) \\ & = I(X_1; X_n) + I(X_2; X_n | X_1) + \ldots + I(X_{n-1}; X_n | X_1, \ldots, X_{n-2}) \\ & = 0 + 0 + \ldots + 1 = 1. \end{aligned}$$

- Let T and F be the top and front side, respectively, of a 6-sided fair die. Compute $I(T; F)$.

$$\begin{aligned} I(T; F) &= \mathrm{H}(T) - \mathrm{H}(T|F) \\ &= \log 6 - \log 4 \\ &= \log 3 - 1. \end{aligned}$$

Section 2

Data Processing (for Markov Chains)

Data processing inequality

Definition 10 (Markov Chain)

$(X, Y, Z) \sim p$ form a Markov chain, denoted $X \to Y \to Z$, if $p_{Z|X,Y}(z|x, y) = p_{Z|Y}(z|y)$, for all $(x, y, z) \in \text{Supp}(X, Y, Z)$.

Example: random walk on a graph.

Claim 11

If $X \to Y \to Z$, then $I(X; Y) \geq I(X; Z)$.

- $$\begin{aligned} \mathrm{H}(Z|X, Y) &= \mathop{\mathrm{E}}_{y \overset{R}{\leftarrow} Y} \left[\mathop{\mathrm{E}}_{x \overset{R}{\leftarrow} X|_{Y=y}} [\mathrm{H}((Z|_{Y=y})|_{X=x})] \right] \\ &= \mathop{\mathrm{E}}_{y \overset{R}{\leftarrow} Y} [\mathrm{H}(Z|_{Y=y})] = \mathrm{H}(Z|Y) \end{aligned}$$
- Hence, $I(X; Z|Y) = \mathrm{H}(Z|Y) - \mathrm{H}(Z|Y, X) = 0$.
- By the chain rule, $I(X; Y, Z) = I(X; Z) + I(X; Y|Z) = I(X; Y) + I(X; Z|Y)$.
- Since $I(X; Y|Z) \geq 0$, we conclude $I(X; Y) \geq I(X; Z)$.□

Fano's Inequality

- How well can we guess X from Y?
- Can be done with no error if $\mathrm{H}(X|Y) = 0$. What if $\mathrm{H}(X|Y)$ is small?

Theorem 12 (Fano's inequality)

For any rvs X and Y, and any (even random) g, it holds that

$$h(P_e) + P_e \log |\mathcal{X}| \geq \mathrm{H}(X|\hat{X}) \geq \mathrm{H}(X|Y)$$

for $\hat{X} = g(Y)$ and $P_e = \Pr\left[\hat{X} \neq X\right]$.

- Note that $P_e = 0$ implies that $\mathrm{H}(X|Y) = 0$
- The inequality can be weakened to $1 + P_e \log |\mathcal{X}| \geq \mathrm{H}(X|Y)$.
- Alternatively, to $P_e \geq \frac{\mathrm{H}(X|Y)-1}{\log|\mathcal{X}|}$
- Intuition for $\propto \frac{1}{\log|\mathcal{X}|}$
- We call $\hat{X}$ an estimator for X (from Y).

Proving Fano's inequality

Let X and Y be rvs, let $\hat{X} = g(Y)$ and $P_e = \Pr\left[\hat{X} \neq X\right]$.

- Let $D = \begin{cases} 1, & \hat{X} \neq X \\ 0, & \hat{X} = X. \end{cases}$

$$\mathrm{H}(D, X|\hat{X}) = \mathrm{H}(X|\hat{X}) + \underbrace{\mathrm{H}(D|X, \hat{X})}_{=0}$$

$$\mathrm{H}(D, X|\hat{X}) = \underbrace{\mathrm{H}(D|\hat{X})}_{\leq \mathrm{H}(D) = h(P_e)} + \underbrace{\mathrm{H}(X|D, \hat{X})}_{\leq P_e \log|\mathcal{X}| \,(?)}$$

- It follows that $h(P_e) + P_e \log|\mathcal{X}| \geq \mathrm{H}(X|\hat{X})$
- Since $X \to Y \to \hat{X}$, it holds that $I(X;Y) \geq I(X;\hat{X})$

  $\implies \mathrm{H}(X|\hat{X}) \geq \mathrm{H}(X|Y)$